

Functions

Functions in Excel are built-in (ready-made) formulas used to perform calculations.

These functions make it easier to perform calculations and analyses data in Excel.

For example, if you are to add values in cells ranging from A1 to A20, a function can be used to reduce the stress of repeating individual cells in the formula.

So instead of =A1+A2+A3+A4+A5.....A20, you can use =SUM (A1:A20) to find the total of values in cell A1 to A20

S/N	Key	Point	Explanation	Example
1	Equal sign	It indicates the start of a formula or function in Excel.	When you type = at the beginning of a cell, Excel understands that you are entering a formula or function, and it will calculate the result based on the expression that follows.	=PRODUCT(B5, A5): with the equal sign beginning this function, excel understands that you want to multiply values of B5 and A5 so it will calculate the result for you.
2	Function name	It is the word you use to tell Excel the kind of calculation or task you want to perform.	Each function name represents a specific action	=SUM (A1:A10): “SUM” is a function name for addition. This function formula will add all values from A1 to A10.
3	Arguments	These are the inputs you provide to the function, such as numbers, cell references, ranges, or text	Some functions require only one argument while others take multiple arguments.	=AVERAGE (M1:M10): A range of cells from M1 to M10 (the numbers for which you want to find the average). =IF (B1 > 50, “Pass”, “Fail”): in this argument, excel will check if B1 is greater than 50 then record pass else it will record fail

Types of Functions

Types	Examples	Explanation
<i>Mathematical Functions</i>	=SUM(B1:B12)	Adds up values in B to B12
	=AVERAGE(B1:B12)	Calculates the average of numbers
	=PRODUCT(B1, B1)	Multiplies values in B1 and B12
	=SQRT(D4)	It will find the square root of the value in cell D4.
<i>Statistical Functions</i>	=MAX(G1:C6)	Gives you the highest number from G1 to G6.
	=MIN(D1:D5)	Shows the smallest number from D1 to D5.
	=COUNT(C1: D12)	This will count all the number of cells (it will count only numbers) specified C1 to D12
<i>Text Functions</i>	=CONCATENATE(A1, " ", B1)	Combines the text in A1 and B1 with a space in between.
	=UPPER(D3)	Changes text from lowercase letters in cell D3 to uppercase letters.
<i>Logical Functions</i>	=IF(D5 > 50, "Pass", "Fail")	Checks if D1 is greater than 50 and returns "Pass" or "Fail".
	=AND(G1 > 0, G1 < 100)	returns TRUE if G1 is between 0 and 100.

<i>Date and Time Functions</i>	=TODAY()	Returns the current date. This function will show today's date.
	=DATEDIF(A1, B1, "D")	It finds the number of days between the dates in A1 and B1. This function calculates the difference between two dates